

Cem Ozcelik

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PROFESSIONAL SUMMARY

Senior Data Scientist with 4+ years of experience translating business problems into measurable ML solutions -- from deep EDA and statistical modelling through production deployment. Industrial Engineering background delivers a native cost-function instinct: before touching architecture, I ask what it costs to be wrong in each direction. Proven full lifecycle ownership across manufacturing analytics, financial risk, and supply chain: **15% increase in critical machine availability** via predictive maintenance, **97% Recall** on medical diagnostic models optimized for asymmetric error cost, and an **8h/week operational overtime reduction** through ML-driven supply chain forecasting. Bilingual Turkish-English with demonstrated ability to communicate insights to both technical and non-technical stakeholders.

CORE COMPETENCIES

Skills: Business Problem Translation, Deep EDA, Statistical Modelling, Anomaly Detection, Predictive Modelling, Data Validation & Quality, Stakeholder Communication

Tools: Python, SQL, XGBoost, Scikit-learn, TensorFlow, PySpark, Docker, Django / FastAPI, PostgreSQL, MSSQL

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe

Jan 2026 - Present

Senior Database Analyst · Maltepe, Istanbul

- Translated supply chain business problems into ML solutions: defined success metrics, built predictive forecasting pipelines, and transferred analytical capabilities to business units -- sustaining an **8-hour weekly reduction in operational overtime**.
- Scaled a Django reporting application into a centralized ML deployment platform -- serving multiple predictive models via RESTful APIs, with a model registry pattern enabling reusable deployment and knowledge sharing across the team.
- Developed clear, actionable insights through dashboards and reports for both technical leads and operations stakeholders, driving data-driven decisions on SKU strategy and supply chain routing.
- Tech:** Python, Django, REST APIs, ML pipelines, Association Rule Mining, PostgreSQL, Docker

Anadolu Pet Akvaryum & Effeffe

Sept 2024 - Dec 2025

IT & Data Analytics Specialist · Maltepe, Istanbul

- Architected and deployed a full-stack internal web application from scratch (Python/Django), establishing the foundational infrastructure for ML model serving and data-driven reporting.
- Engineered predictive supply chain optimization models using Python and PL/SQL, improving inventory turnover and reducing stockouts through data-validated forecasting.
- Conducted Route-Level Profitability Analysis with advanced SQL -- defined cost-to-profit ratios that delivered a **3% improvement in overall logistics efficiency**.
- Tech:** Python, Django, PL/SQL, Oracle, SOLID principles, full-stack deployment

Birinci Automotive

Jan 2022 - Sept 2024

Mid-Level Data Analyst · Cayirova, Kocaeli

- Conducted deep EDA on years of sensor failure logs and machine breakdown records -- mapping breakdown signatures against time-series features to identify candidate predictors for a predictive maintenance model, delivering a **15% weekly increase in critical machine availability** and **15% OEE improvement**.
- Built a full ML pipeline from data extraction (MSSQL ETL) through feature engineering on machine sensor data, unsupervised anomaly detection modelling, and operational deployment -- replacing reactive, manual workflows.
- Defined data quality and validation practices for sensor ingestion pipelines; trained shift supervisors on interpreting model alerts, driving operational adoption of ML-based decision making.
- Tech:** Python, MSSQL, ETL pipelines, unsupervised clustering, anomaly detection, time-series analysis

Junior Data Analyst · Cayirova, Kocaeli

- Applied EDA and statistical methods to workforce datasets to uncover hidden patterns supporting data-driven HR policy decisions.
- Designed automated data ingestion pipelines, **improving reporting efficiency by 15%**.
- **Tech:** Python, SQL, statistical analysis, ETL

PROJECTS**End-to-End Credit Risk Prediction -- MLOps & Full Lifecycle** github.com/ccemozclk/Credit-Risk

Complete ML solution covering the full lifecycle: deep EDA on raw financial data, feature engineering, class-weighted XGBoost optimized for heavy imbalance (63% F1, temporal holdout), and an interactive Django EDA platform for non-technical stakeholders -- fully containerized with Docker. Architecture pattern transfers directly to user behavior anomaly detection and risk modelling.

Python · XGBoost · Django · Docker · Feature Engineering · Interactive EDA Platform

Scalable Recommendation Engine -- Personalization at Distributed Scale github.com/ccemozclk/Spark-Recommendation-Engine

End-to-end personalized recommendation system on Apache Spark -- directly analogous to sports platform personalization. K-Means behavioral segmentation + ALS collaborative filtering on Databricks. Tuned for conversion-rate business KPI, not just RMSE. Addresses data skew and Spark-specific scaling challenges at distributed scale.

Python · PySpark · Apache Spark · Databricks · K-Means · ALS · Distributed Computing

AI-Powered Medical Diagnostics -- Cost-Function Optimized ML github.com/ccemozclk/Medical-Diagnostics

End-to-end CNN diagnostic system for pneumonia detection from chest X-rays. Achieved **97% Recall** by explicitly optimizing for asymmetric error cost (False Negative >> False Positive). Grad-CAM heatmaps for model explainability -- identified and corrected a dataset bias invisible without visualization. Transfer Learning on VGG16/ResNet50 for imbalanced medical datasets.

Python · TensorFlow-Keras · CNN · Transfer Learning · Grad-CAM · Explainable AI

EDUCATION**Kirikkale University -- B.Sc. Industrial Engineering**

2016 - 2021

Rigorous foundation in statistical analysis, linear algebra, and optimization. MILP with IBM ILOG CPLEX; Multi-Criteria Decision Making (AHP, TOPSIS). Operations research principles applied directly to ML cost-function design and business problem translation.

CERTIFICATIONS

DataCamp -- Associate Data Scientist Career Track (PySpark, Feature Engineering, Recommendation Engines)

2024

Miuul -- Data Science & Machine Learning Bootcamp (end-to-end ML lifecycle, 10+ real-world projects)

2022

SKILLS

Core Languages: Python (Advanced), SQL (Advanced), Bash

ML & Statistical Modelling: XGBoost, Scikit-learn, TensorFlow, PyTorch, CNNs, Explainable AI (Grad-CAM), Transfer Learning, h2o.ai

Big Data: Apache Spark (PySpark), Databricks, Distributed Computing

Deployment & MLOps: Docker, Django, FastAPI, Flask, REST APIs, Git, Linux, End-to-End ML Deployment

Data Engineering: ETL Pipelines, PostgreSQL, MSSQL, Oracle, Pandas, NumPy, Interactive EDA platforms (Django)

Cloud: AWS (EC2, S3)

Languages: Turkish (Native), English (Fluent), Spanish (Basic)